

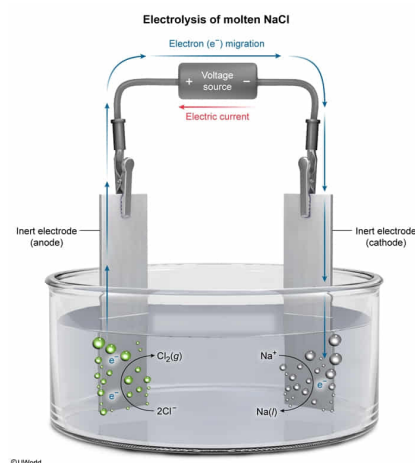
Which statements correctly describe electrolysis? During electrolysis:

- I. an electrochemical decomposition reaction occurs.
 - II. electric current is supplied as the energy to perform an endergonic reaction.
 - III. the electrochemical cell operates with a negative ΔG .
 - IV. electric current flows in a nonspontaneous direction within an electrochemical cell.
- ☐ A. I and IV only [6%]
☐ B. I, II, and III only [16%]
☐ C. II and IV only [23%]
☒ D. I, II, and IV only [53%]

Correct

53% answered correctly

Explanation:

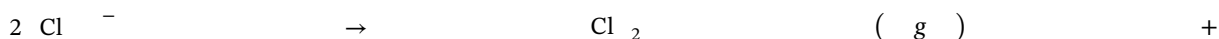


Using a loosely applied analogy, an **electrolytic cell** can be conceptualized as a **galvanic (voltaic) cell** that is operating in reverse. Although the electrolytic cells may be constructed differently (one chamber vs. two) and some chemical reactions are irreversible, accounting for these distinctions and limitations, electrolytic and galvanic cells essentially perform in an opposite manner.

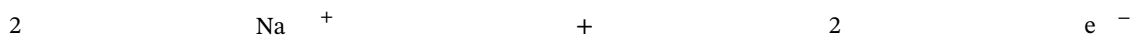
In a galvanic cell, a spontaneous, **exergonic** chemical reaction ($\Delta G < 0$) produces electric current. Conversely, in an electrolytic cell, electric current from an external source (such as a battery) is supplied to drive a nonspontaneous, **endergonic** chemical reaction ($\Delta G > 0$) within the **electrochemical cell**.

In both galvanic and electrolytic cells, **reduction** takes place at the **cathode** and oxidation takes place at the anode, but the role of a given electrode is opposite in each cell type. The electrode functioning as the anode in a galvanic cell functions as the cathode in an electrolytic cell (and vice versa). As such, the direction of electrical current is reversed in an electrolytic cell compared with a galvanic cell. (Recall that, by convention, the direction of the **electric current** is taken to be opposite to the direction of electron migration.)

The operation of an electrolytic cell causes electrolysis, which is the **nonspontaneous**, electrochemical **decomposition** of a compound (**Number 1**). At the electron-deficient anode, electron-rich species (such as anions) are oxidized. For example, in the electrolysis of molten NaCl, Cl^- is oxidized to Cl_2 gas:



At the electron-rich cathode, electron-deficient species (such as metal cations) are reduced. In the example of the electrolysis of molten NaCl, Na^+ is reduced to Na metal:



The net reaction for the electrochemical process gives the overall nonspontaneous, decomposition reaction:



Because electrolysis uses an electrolytic cell, the supplied electrical current flows in a nonspontaneous direction within the

electrochemical cell (**Number IV**), which supplies the energy needed to perform a nonspontaneous (endergonic) decomposition reaction (**Number II**).

(**Number III**) Electrolysis uses an *electrolytic* cell that has a *positive*, not a negative, ΔG .

Educational objective:

In an electrolytic cell, electric current from an external source is supplied to provide the necessary energy to perform a nonspontaneous (endergonic), decomposition reaction of a compound within the electrochemical cell.

Time spent:0 sec

Subject:
General Chemistry

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System:
Solutions and Electrochemistry